

Unit 3: Knowledge Representation and Probabilistic Reasoning

SubUnit 3.6: Fuzzy Logic - Membership Functions and Inference Systems

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From Classical to Fuzzy Logic

Classical (Binary) Logic:

- Everything is either TRUE (1) or FALSE (0)
- Sharp, crisp boundaries
- Example: $Temperature > 25^{\circ}C \Rightarrow Hot = TRUE$

Problem: Real World is Vague!

- Is $25.1^{\circ}C$ hot? What about $24.9^{\circ}C$?
- "Tall person", "high temperature", "fast speed"
- Human reasoning involves degrees and approximations

Fuzzy Logic Solution

Introduced by Lotfi Zadeh (1965): Allows **partial truth** - values between 0 and 1. Represents the **degree of membership** to a set.

Example

Temperature = $26^{\circ}C$: Hot = 0.7, Warm = 0.3, Cold = 0.0 (70% hot, 30% warm, 0% cold)

Fuzzy Sets: Basic Concept

Classical Set:

- Element either belongs (1) or doesn't belong (0)
- Characteristic function: $\chi_A(x) \in \{0, 1\}$

Fuzzy Set:

- Element can have **partial membership**
- Membership function: $\mu_A(x) \in [0, 1]$
- $\mu_A(x) = 1$: Full membership
- $\mu_A(x) = 0$: No membership
- $0 < \mu_A(x) < 1$: Partial membership

Fuzzy Set: "Tall"

Height 190cm $\rightarrow \mu_{tall} = 1.0$

Height 180cm $\rightarrow \mu_{tall} = 0.7$

Height 170cm $\rightarrow \mu_{tall} = 0.3$

Height 150cm $\rightarrow \mu_{tall} = 0.0$

Classical Set: "Tall"

Height ≥ 180 cm $\rightarrow$ Tall = 1

Height < 180 cm $\rightarrow$ Tall = 0

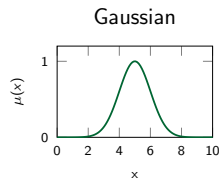
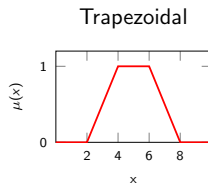
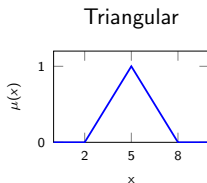
Notation: Fuzzy set $A =$

$\{(x, \mu_A(x)) | x \in X\}$

Reference: Rich et al. (2009), Section 8.4.1

Membership Functions: Common Types

Definition: $\mu_A : X \rightarrow [0, 1]$ maps domain values to degree of membership



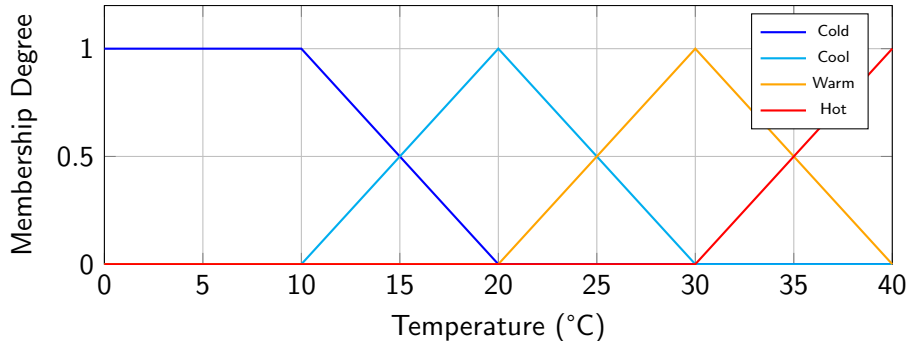
Membership Function Formulas:

- **Triangular:** $\mu(x) = \max\left(0, \min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right)\right)$
- **Trapezoidal:** Extension of triangular with flat top
- **Gaussian:** $\mu(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$
- **Sigmoid:** $\mu(x) = \frac{1}{1+e^{-k(x-c)}}$

Reference: Rich et al. (2009), Section 8.4.2

Example: Temperature Fuzzy Sets

Linguistic Variables: "Cold", "Cool", "Warm", "Hot"



Example Evaluation at 28°C:

• $\mu_{Cold}(28) = 0.0$, $\mu_{Cool}(28) = 0.2$, $\mu_{Warm}(28) = 0.8$, $\mu_{Hot}(28) = 0.0$

Reference: Rich et al. (2009), Section 8.4

Fuzzy Set Operations

Let **A** and **B** be fuzzy sets with membership functions $\mu_A(x)$ and $\mu_B(x)$

1. Union (OR):

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

2. Intersection (AND):

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$$

3. Complement (NOT):

$$\mu_{\neg A}(x) = 1 - \mu_A(x)$$

Alternative T-norm and T-conorm operators:

- **Product:** $\mu_{A \cap B}(x) = \mu_A(x) \times \mu_B(x)$
- **Probabilistic Sum:** $\mu_{A \cup B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x) \times \mu_B(x)$

Example

If $\mu_{Hot}(30) = 0.7$ and $\mu_{Humid}(30) = 0.6$

Hot OR Humid:

$$\max(0.7, 0.6) = 0.7$$

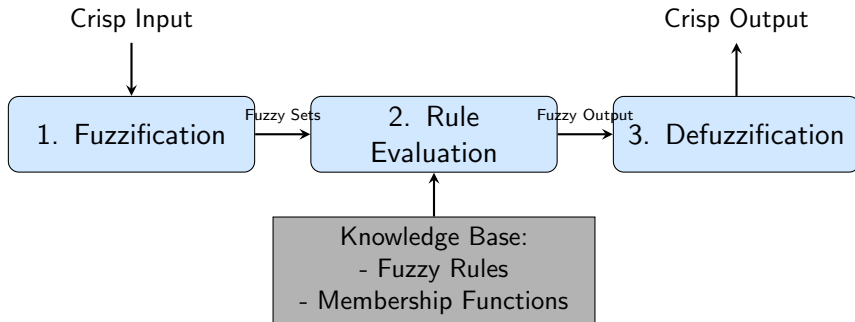
Hot AND Humid:

$$\min(0.7, 0.6) = 0.6$$

NOT Hot:

$$1 - 0.7 = 0.3$$

Fuzzy Inference System (FIS): Architecture



Three Main Steps:

- ➊ **Fuzzification:** Convert crisp inputs to fuzzy values using membership functions
- ➋ **Rule Evaluation:** Apply fuzzy IF-THEN rules from knowledge base
- ➌ **Defuzzification:** Convert fuzzy output to crisp value

Reference: Russell & Norvig (2020), Section 13.7

Step 1: Fuzzification

Goal: Convert crisp input values to degrees of membership in fuzzy sets

Example: Temperature Control System

Inputs:

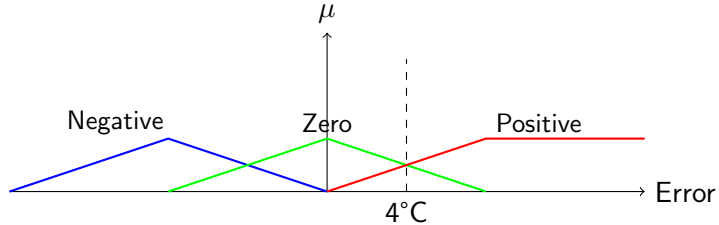
- Current Temperature = 18°C
- Desired Temperature = 22°C

Calculate Temperature Error: $\text{Error} = 22 - 18 = 4^\circ\text{C}$

Fuzzify Error (using membership functions):

- Error = 4°C
- $\mu_{\text{Negative}}(4) = 0.0$ (Not negative)
- $\mu_{\text{Zero}}(4) = 0.2$ (Slightly zero)
- $\mu_{\text{Positive}}(4) = 0.8$ (Mostly positive)

Example



Reference: Rich et al. (2009), Section 8.4.4

Step 2: Fuzzy IF-THEN Rules

Format: IF [antecedent] THEN [consequent]

Fuzzy Rules for Temperature Control:

- ① IF error is **Negative** THEN heater is **Off**
- ② IF error is **Zero** THEN heater is **Low**
- ③ IF error is **Positive** THEN heater is **High**

Rule Evaluation (using min operator):

- Rule 1: $\min(\mu_{Negative}(4), \mu_{Off}) = \min(0.0, 1.0) = 0.0$
- Rule 2: $\min(\mu_{Zero}(4), \mu_{Low}) = \min(0.2, 1.0) = 0.2$
- Rule 3: $\min(\mu_{Positive}(4), \mu_{High}) = \min(0.8, 1.0) = 0.8$

Multiple Antecedents:

- IF error is Positive **AND** rate is High THEN heater is Medium
- Use min for AND: $\min(\mu_{error}, \mu_{rate})$
- Use max for OR: $\max(\mu_{error}, \mu_{rate})$

Note

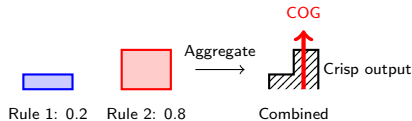
Each rule fires with a strength equal to its antecedent's membership degree

Fuzzy Inference: Mamdani Method

Most Common Method for Fuzzy Control

Steps:

- 1 Evaluate each rule's antecedent (using min/max)
- 2 Apply implication (clip consequent membership function)
- 3 Aggregate all rule outputs (using max)



Mamdani Characteristics:

- Intuitive and widely used
- Output is a fuzzy set (requires defuzzification)
- Good for systems with multiple outputs
- Computationally more expensive

Step 3: Defuzzification

Goal: Convert fuzzy output to single crisp value **Common Defuzzification Methods:**

1. Center of Gravity (Centroid) - Most Popular:

$$x^* = \frac{\int x \cdot \mu(x) dx}{\int \mu(x) dx}$$

Discrete version: $x^* = \frac{\sum x_i \cdot \mu(x_i)}{\sum \mu(x_i)}$

2. Mean of Maximum (MOM):

- Average of all values with maximum membership

3. Bisector:

- Value that splits area under curve in half

Example: COG Calculation

Output membership: (50, 0.2), (60, 0.5), (70, 0.8), (80, 0.5), (90, 0.2)

$$x^* = \frac{50(0.2)+60(0.5)+70(0.8)+80(0.5)+90(0.2)}{0.2+0.5+0.8+0.5+0.2} = \frac{154}{2.2} = 70C$$

Reference: Rich et al. (2009), Section 8.4.7

Complete Example: Fuzzy Tipping System

Problem: Determine tip based on service quality and food quality

Inputs:

- Service quality (0-10 scale)
- Food quality (0-10 scale)

Output: Tip percentage (0-25%)

Fuzzy Rules:

- 1 IF service is **Poor** THEN tip is **Low**
- 2 IF service is **Good** THEN tip is **Medium**
- 3 IF service is **Excellent** OR food is **Delicious** THEN tip is **High**

Complete Example: Fuzzy Tipping System

Scenario: Service = 7, Food = 8

Fuzzification:

- $\mu_{Good}(7) = 0.6$, $\mu_{Excellent}(7) = 0.4$
- $\mu_{Delicious}(8) = 0.7$

Rule Evaluation:

- Rule 2: $\min(0.6, 1.0) = 0.6 \rightarrow$ Medium tip with strength 0.6
- Rule 3: $\max(0.4, 0.7) = 0.7 \rightarrow$ High tip with strength 0.7

Result: After defuzzification $\rightarrow$ Tip 18%

Reference: Russell & Norvig (2020), Example 13.7

Applications of Fuzzy Logic

Control Systems:

- **Industrial Control:** Manufacturing processes, robotics
- **Automotive:** Anti-lock braking (ABS), transmission control, cruise control
- **Home Appliances:** Washing machines, air conditioners, rice cookers
- **Elevators:** Smooth acceleration/deceleration

Decision Support:

- **Medical Diagnosis:** Interpreting symptoms with uncertainty
- **Financial Trading:** Risk assessment, portfolio management
- **Pattern Recognition:** Image processing, handwriting recognition

Other Applications:

- **Natural Language Processing:** Understanding vague terms
- **Expert Systems:** Incorporating human expert knowledge
- **Robotics:** Navigation in uncertain environments
- **Consumer Electronics:** Camera autofocus, video stabilization

Success Story

Sendai subway system (Japan) uses fuzzy logic for smooth train control - saves energy and improves passenger comfort

Reference: Rich et al. (2009), Section 8.4.8

Fuzzy Logic: Strengths and Weaknesses

Advantages:

- **Natural:** Models human reasoning
- **Flexible:** Handles imprecise data
- **Simple:** Easy to understand and implement
- **Robust:** Tolerates noisy inputs
- **Nonlinear:** Models complex systems
- **Linguistic:** Uses natural language rules
- **Practical:** Proven in many applications

Limitations:

- **No learning:** Rules must be specified manually
- **Tuning:** Membership functions need adjustment
- **Validation:** Hard to verify correctness
- **Not probabilistic:** Different from Bayesian reasoning
- **Scalability:** Many rules for complex systems
- **No standard:** Multiple inference methods

Fuzzy Logic vs. Probability:

Aspect	Fuzzy Logic	Probability
Measures	Degree of membership	Likelihood of event
Question	How much?	How likely?
Example	"Tall" (degree 0.7)	$P(\text{Rain}) = 0.7$
Complement	$\mu_{\text{not}}(x) = 1 - \mu(x)$	$P(\neg A) = 1 - P(A)$

Reference: Russell & Norvig (2020), Section 13.7

Summary: Fuzzy Logic

Key Concepts:

- **Fuzzy Sets:** Partial membership (0 to 1) instead of binary
- **Membership Functions:** Define degree of belonging (triangular, trapezoidal, Gaussian)
- **Linguistic Variables:** Natural language terms ("hot", "cold", "medium")
- **Fuzzy Operations:** AND (min), OR (max), NOT ($1-x$)

Fuzzy Inference System (FIS):

- 1 **Fuzzification:** Crisp input $\rightarrow$ Fuzzy values
- 2 **Rule Evaluation:** Apply IF-THEN rules
- 3 **Defuzzification:** Fuzzy output $\rightarrow$ Crisp value (COG, MOM)

When to Use Fuzzy Logic:

- Systems with vague, imprecise, or linguistic information
- Expert knowledge available but hard to formalize mathematically
- Control systems requiring smooth, continuous outputs
- When human-like reasoning is desired